

AlphaPilot AI — Institutional Options Trading Copilot

Comprehensive Hackathon & Technical Presentation
Document

- 📄 **Project Tagline:** An institutional-grade options trading terminal powered by Explainable AI, dynamic contract scanning, and automated multi-layered risk guards.



Executive Summary & Market Challenge

Retail options trading suffers from significant systemic inefficiencies. Individual traders frequently experience high emotional stress, leading to poor execution discipline, lack of automated risk management, and over-reliance on "black-box" decision models. AlphaPilot AI bridges this gap by offering a transparent, rule-backed trading copilot integrated directly with live market feeds and execution APIs.

Market Challenge (The Problem) vs. AlphaPilot AI Solution

Market Challenge (The Problem)	AlphaPilot AI Solution
Emotional & impulsive manual trade execution	Deterministic, rule-backed AI trade signal generation
Lack of real-time multi-factor risk controls	Automated, multi-layered risk guards with hard locking
Opaque "black box" AI predictions	Full transparency via Explainable AI breakdown and Evidence logs
Poor position management under volatility	Direct Alpaca Paper Trading API integration & Emergency Exit controls

System Architecture & Workflow Pipeline

The platform follows a strict, sequential operational pipeline to evaluate market opportunities while protecting capital:

01

Refresh Market Data

Real-time market telemetry is fetched via Alpaca and IEX data feeds.

02

Run AI Analysis

The engine evaluates technical indicators (SMA20, SMA50, RSI, MACD, Volume) against deterministic threshold rules.

03

Scan Option Chains

Filters contract candidates dynamically based on Days to Expiration (DTE), Open Interest, and Pricing.

04

Run Risk Guards Check

Applies hard risk parameter checks including position limits, account risk, and daily drawdown caps.

05

Order Execution Routing

Orders are securely transmitted to the Alpaca API only when all previous verification gates pass.

Explainable AI Engine & Signal Evidence

AlphaPilot AI eliminates black-box uncertainty. Every decision is fully audited through explicit technical checks:

- **Price vs. Moving Averages**
Verifies whether underlying price sits above key SMA levels (SMA20 / SMA50).
- **Momentum Verification**
RSI values are checked for optimal entry zones, avoiding overbought or oversold extremes.
- **Trend Crossover**
MACD indicator bullish/bearish alignment.
- **Volume Confirmation**
Ensures underlying volume supports price momentum.
- **Confidence Threshold**
Requires a minimum 70% technical pass rate. Signals below this threshold default to a safe "NO TRADE" state.

Algorithmic Option Selection Engine

When technical criteria are satisfied, the contract scanner evaluates option chains to select the optimal contract:

Parameter Filtering Logic & Criteria

Parameter	Filtering Logic & Criteria
Days to Expiration (DTE)	Target optimal liquidity and delta window (typically 15–30 DTE)
Liquidity Metrics	Evaluates Open Interest and daily trading volume thresholds
Selection Score	Custom algorithmic scoring model prioritizing optimal risk-reward ratios
Contract Surfacing	Selects and displays the single highest-scoring strike candidate



Hardened Multi-Layered Risk Guards

Risk management is built directly into the core system architecture. No trade can bypass these mandatory safety barriers:

Position Sizing Limit Guard

Caps position size based on overall account portfolio allocation.

Maximum Account Risk Threshold

Prevents over-leveraging on single contract trades.

Daily Drawdown Cap Check

Automatically locks trading if daily loss limits are reached.

Duplicate Order Guard

Prevents redundant entries into identical contract positions.

Execution, Performance Analytics & System Health

AlphaPilot AI provides transparent auditing and performance reporting:

Alpaca Paper Integration

Real-time order routing with order fill logging, price history, and execution timestamps.

Position Manager

Tracks active SPY options, closed trades, and features a one-click Emergency Exit All Positions override.

Performance Analytics

Calculates real Win Rates, Realized P&L, and trade-by-trade cumulative breakdowns with zero simulated data.

System Health Status

Real-time status indicators confirming active connections to Alpaca APIs, IEX Market Data feeds, and auto-refresh mechanisms.